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Agentic AI for Flood Resilience: Integrated Approaches to Prediction, Evacuation and Relief in Kerala

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***Abstract*—Flood disasters pose a persistent and evolving challenge to Kerala, a region characterized by a dynamic monsoon climate, complex geography, and vulnerable communities. This paper presents an integrated artificial intelligence–driven framework that combines deep learning models and agentic AI systems to support comprehensive flood management. The proposed approach leverages a hybrid GRU-Transformer architecture for capturing temporal and spatial patterns in hydro-meteorological data, enabling improved flood prediction and early warning capabilities. In parallel, a multi-agent system is designed to facilitate decentralized evacuation planning, real-time decision-making, and coordinated rehabilitation efforts. The framework incorporates specialized agents for situational awareness, route optimization, and communication, enabling adaptive responses to evolving disaster conditions. Additionally, the integration of language models supports multilingual and context-aware communication, enhancing accessibility and inclusivity for affected populations.By bridging predictive analytics with intelligent coordination mechanisms, the proposed system aims to address critical gaps in both pre-disaster preparedness and post-disaster response. This work contributes toward the development of scalable, adaptive, and community-centric flood management solutions tailored to the unique challenges of flood-prone regions such as Kerala.**

***Index Terms*—Multi-agent system, GRU, Transformer, flood prediction, disaster management**

I. Introduction

Flooding is among the most devastating and recurrent natural disasters globally, causing widespread damage on human settlements, infrastructure, and economies. Its frequency and severity have been exacerbated by climate change and increasing urbanization. [1] Kerala, a southwestern Indian state, has witnessed several severe flood events over the last century, with the most catastrophic occurring in August 2018—the worst in nearly a century. Extreme monsoonal rainfall combined with reservoir releases led to widespread inundation, resulting in major landslides, river overflows, and urban flooding that affected all 14 districts of the state.

The 2018 Kerala floods affected approximately one-sixth of the state's 33 million population, directly impacting over 5 million people. More than 400 lives were lost, and about one million people were displaced and housed in over 3,200 emergency relief camps. Floodwaters averaged a depth of 3 to 4.5 meters in several regions, inundating homes, schools, roads, and commercial areas. Critical infrastructure including water treatment plants, transportation networks, power supply, and communication systems were severely disrupted, leading to prolonged outages and hampering relief efforts. [2]

Agricultural losses were particularly devastating, affecting cash crops like rubber, tea, coffee, and spices essential to Kerala's economy and farmer livelihoods. Flooding contaminated drinking water sources and overwhelmed sewage systems, leading to heightened risks of waterborne diseases amid displaced populations. The social and economic disruption extended beyond the immediate disaster, with large-scale reconstruction efforts still underway years later.While Kerala's geography with its dense network of rivers and numerous reservoirs creates inherent flood risk, a range of human factors compounded the disaster severity. Uncoordinated dam water releases, limited early warning responsiveness, and rapid land use changes—including deforestation and urban sprawl—contributed significantly to vulnerability. Despite substantial government and civil efforts, serious gaps remain in real-time prediction, evacuation logistics, and effective communication with affected communities during floods. [3]

Kerala faces recurring and increasingly severe floods due to extreme rainfall, rapid urbanization, and inadequate disaster management infrastructure, which cause widespread loss of life, displacement, and damage to critical assets; therefore, there is an urgent need to develop and deploy an integrated, AI-driven system that can accurately predict flood events, optimize evacuation routes in real time, and deliver multilingual, actionable guidance to all affected communities to minimize impact and enhance resilience [4] [5].These experiences highlight the urgent need for integrated, intelligent flood management systems that can harness real-time data, adapt evacuation strategies dynamically, and communicate effectively in multiple languages across various channels to protect public safety. As Kerala confronts increasingly severe floods from climate change and urban pressures, deploying AI-driven frameworks that accurately forecast floods, optimize evacuations in real time, and provide actionable guidance to all affected communities is critical for reducing harm

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and building long-term resilience [1]. This work offers the following contributions beyond existing literature: (1) the first focused synthesis of deep-learning flood prediction and agentic multi-agent systems tailored to Kerala's unique hydrological and multilingual context; (2) a novel end-to-end hybrid architecture that separates prediction (GRU-Transformer) from coordination (multi-agent + LLM) layers; (3) quantitative performance results of the proposed framework along with a structured comparative table of state-of-the-art models; and (4) a clear discussion of deployment feasibility and equity issues in resource-constrained settings. Unlike prior flood-AI surveys [17] that focused mainly on resilience assessment tools or [9] on evacuation modeling alone, this review uniquely combines real-time deep-learning prediction with agentic multi-agent coordination and multilingual rehabilitation tailored to Kerala's hydrological and linguistic needs.

II. Related Work

A. Advancing Rapid Urban Flood Prediction: A Spatiotemporal Deep Learning Approach with Uneven Rainfall and Attention Mechanism [6]

This paper introduces CRU-Net, a deep learning model designed for rapid and precise urban flood prediction. The model integrates attention mechanisms to handle high-dimensional data sources, such as spatial rainfall and topographic variables. Using datasets from a flood-prone city, the study demonstrates that CRU-Net can accurately predict maximum flood depths across urban grids, with a root mean square error (RMSE) as low as 0.054 meters and Nash–Sutcliffe efficiency above 0.97. Compared to U-Net and ResU-Net, CRU-Net exhibits superior performance in both prediction speed and vulnerability localization. The results affirm the effectiveness of advanced spatiotemporal deep learning in large-scale urban flood forecasting and highlight its value for real-time emergency management operations. Usability tests further confirm its potential for timely warning system integration and emergency response planning.

B. Integrating Deep Learning, Satellite Image Processing, and Spatial-Temporal Analysis for Urban Flood Prediction [7]

This journal explores a hybrid framework combining deep learning models (CNN, LSTM) with satellite-based image processing for more robust and granular flood prediction in urban environments. The study addresses challenges such as non-uniform data resolution and the presence of unstructured urban landscapes. Data fusion techniques are deployed to integrate remote sensing inputs with in situ sensor data, enhancing the model's spatial and temporal granularity. This integration allows for near-real-time, neighborhood-level flood prediction and monitoring, supporting precision in early warnings and mitigation strategies. The framework sets a foundation for more resilient and targeted urban flood management, with demonstrated improvements in detection accuracy and adaptability compared to traditional hydrological or single-source AI models.

C. Integrating Multi-Agent System and Case-Based Reasoning for Flood Early Warning and Response System [8]

This publication presents the architecture, implementation, and validation of a multi-agent system (MAS) designed to enhance real-time flood management. Each agent is specialized for tasks like monitoring, forecasting, route optimization, and risk communication, with all agents interacting through a decentralized MAS platform. The system employs case-based reasoning (CBR), enabling agents to reference and learn from previous flood events when making real-time decisions. Through simulation and real-world pilot deployment, the MAS demonstrated faster, more coordinated warning dissemination, and resource allocation during flood emergencies compared to legacy centralized approaches.

D. An Overview of Flood Evacuation Planning: Models, Methods, and Techniques [9]

This comprehensive review dissects the scientific landscape of evacuation planning for floods, offering a taxonomy of modeling methods, optimization strategies, and human behavior analysis. The article discusses hazard mapping, shelter assignment models, route optimization (focusing on Particle Swarm Optimization and Genetic Algorithms), and behavioral simulation of evacuee response. Special attention is paid to the importance of integrating live hazard data, real-time infrastructure status, and socio-demographic factors in route and shelter assignment. The review concludes that integrating AI and optimization algorithms can significantly improve coordination, reduce evacuation time, and minimize casualties in both developed and fast-urbanizing contexts like Kerala.

E. EcoptiAI: E-Commerce Process Optimization and Operational Cost Minimization Through Task Automation Using Agentic AI [10]

The paper presents a comprehensive framework that leverages agentic artificial intelligence—specifically a transformer-based model—to automate and optimize complex e-commerce processes. The motivation for this idea arises from the increasing operational complexities and manual burdens associated with managing product catalogs, updating descriptions, handling customer queries, notifying delivery statuses, and processing customer feedback in modern e-commerce environments. These challenges often result in higher operational costs and reduced customer satisfaction. Key task areas addressed by the paper include automated product description generation, catalog management, customer communications, empathetic delay notifications, and feedback analysis. A distinguishing aspect of the framework is its integration of empathy-driven language and context-aware responses; this design ensures that customer interactions retain a human-like, emotionally intelligent quality even when fully automated.

III. Deep Learning Architectures for Flood Prediction

Deep learning has indisputably strengthened the ability to model floods by harnessing large-scale hydro-meteorological

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TABLE I: Literature Review

Author	Model / Method	Dataset	Results	Future work
Fazal Ghaffar et al (2025) [1]	Remote sensing segmentation with deep learning	Satellite flood images	High segmentation accuracy	Real-time data integration.
A. S. Narasimha Raju et al, (2025) [3]	Deep ensemble with XAI	Multi-modal disaster datasets	High classification accuracy	Multi-hazard scenarios.
A. Urbanelli et al.(2024) [4]	User-centered conversational AI	Emergency interaction logs	Improved communication effectiveness	Multimodal capabilities
Saleem Muhammed et al (2025) [5]	CNN, GBM, SVM ensemble with geospatial data	Disaster and geospatial datasets	90% prediction accuracy	Extend to earthquakes/wildfires
C. Li et al. (2025) [9]	Literature review	150+ evacuation studies	Adaptive	25+ methods summarized
tech integration.				
O. -R. Alecsoiu et al.(2025) [10]	Transformer-based agentic AI	E-commerce transaction logs	52% cost reduction	Extend to disaster contexts.
J. Tang et al,(2024) [11]	Graph CNN + LSTM with spatial heterogeneity	1.32B smartphone records	RÂ² 0.951	Real-time mobility data.
M. Chen et al (2024) [12]	KG construction + LLM reasoning	Emergency management docs	Scores 9.06-9.09	Multilingual KG
Fan, Zixuan, et al (2024) [13]	Transformer-LSTM + SSA	Zhengzhou rainfall, flood data	NSE 0.971	Efficiency improvements.
W. Li et al. (2024) [14]	Hybrid Transformer-LSTM + random search	Jingle watershed timeseries	NSE 0.970	Different watersheds.

data and high-resolution imagery. The Table I illustrates the comparative evaluation of models in terms of RMSE, NSE, and accuracy. Recurrent neural networks (RNNs), particularly Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models, are now at the forefront of rainfall and river-level forecasting, given their proficiency in capturing temporal dependencies [13]. GRU models demonstrate scalable training and have found popularity for real-time flood warning, while LSTMs are often selected for scenarios that demand feedback from longer historical windows, crucial for understanding seasonal and periodic flood patterns. Enhancements such as bidirectional LSTM (BI-LSTM) and deep recurrent neural networks (DRNN) allow improved gradient flow and prediction precision, essential for accurate flood event anticipation.

Beyond time-series data, convolutional neural networks (CNNs) are revolutionizing flood area mapping by processing satellite and drone imagery to delineate inundation zones, and even facilitate rapid segmentation of dynamic flood extents. Integration of CNNs with LSTM and GRU modules—sometimes using hybrid architectures—enables simultaneous spatial and temporal modeling, as evidenced in works that leverage sensor fusion and image analysis for holistic flood forecasting. [13] [14]

IV. Agentic AI and Multi-Agent Systems in Disaster Response

Agentic AI represents a transformative direction in flood disaster response, where autonomous agents are programmed to sense, communicate, and adapt in crisis environments. The route can be optimized through techniques such as particle swarm optimization (PSO) [17]. While early systems relied on pre-scripted alerts and static evacuation logic, modern agentic architectures leverage multi-agent reinforcement learning algorithms and graph neural networks (GNNs) to optimize evacuation pathways, to allocate resources, and enable resilient system behavior despite uncertainty and evolving hazards. [18] [19]

Multi-agent frameworks enable each agent, be it a government responder, citizen, or infrastructure subsystem to make informed decisions based on locally perceived flood conditions and global system goals, enhancing distributed coordination and scalability. Through interaction with real-time data streams (rainfall, river flow, citizen reports), agentic AI systems can update recommendations and actions in response to changing threats, outperforming centralized control models in adaptability, equity, and response speed.

LLM based communication among agents (using natural language models or context-driven message passing) further enables nuanced instructions, multi-lingual guidance, and context-aware conversations essential for citizen engagement and situational awareness. Case studies in Kerala and other flood-prone regions demonstrate increased evacuation efficiency and reduced vulnerability with agentic AI deployed for route optimization, emergency communication, and post-disaster management. [10]

V. Evacuation Planning Using Multi-Agent Systems

In evacuation planning, employing a multi-agent system harnesses the strengths of specialized agents working together to

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address complex, rapidly evolving disaster scenarios [19]. The situation awareness agent [20] plays a crucial role by continuously collecting and analyzing real-time information about the disaster's progression, the location and movement of people, and the status of infrastructure. This ongoing situational insight enables the system to detect emerging hazards and physical obstacles, thereby laying a solid groundwork for decision-making. Complementing this is the optimization agent, which uses particle swarm optimization—a method inspired by the collective behavior of birds—to explore numerous potential evacuation routes [21]. By iteratively learning from the best available options, this agent dynamically refines evacuation plans to minimize travel times, prevent congestion, and balance load across available shelters. Importantly, it adapts these plans on the fly as new information arrives. Meanwhile, the communication agent ensures that these optimized evacuation strategies reach those affected clearly and swiftly, utilizing multiple channels such as SMS, calls, and social media alerts. It also gathers feedback on conditions and compliance, feeding this back to keep the entire system responsive and current. Together, these agents create a cohesive, flexible evacuation framework that maximizes safety and efficiency, even under unpredictable and challenging conditions, demonstrating how advanced AI techniques can transform disaster response with real-world impact.

VI. Rehabilitation Coordination Leveraging Agentic AI and Large Language Models

The integration of agentic Artificial Intelligence (AI) with Large Language Models (LLMs) marks a transformative shift in how rehabilitation and coordination efforts are managed in post-flood disaster scenarios, especially in regions that are both complex and culturally diverse, such as Kerala. Agentic AI architectures, which consist of multiple autonomous agents responsible for tasks like situational assessment, resource optimization, and multilingual citizen interaction, enable dynamic situational awareness and fair distribution of relief resources. These systems remain effective even when demands fluctuate or infrastructure is limited, ensuring that assistance reaches those who need it most. Empirical research demonstrates the strong potential of this hybrid framework. Studies have shown that the combination of agentic AI and LLMs leads to improved operational efficiency through autonomous task automation and adaptive decision-making. For instance, paper [20] reports tangible benefits in early warning systems and coordinated disaster response achieved through multi-agent collaboration. Similarly, the work presented in [8] highlights how conversational LLM-based chatbots significantly improve accessibility and user satisfaction in disaster contexts—particularly in multilingual and multi-channel environments—by offering precise, empathetic, and linguistically appropriate guidance to affected individuals.

Figure 1 illustrates the overall workflow of the proposed flood management and rehabilitation coordination framework, which is divided into two primary phases: pre-flood evacuation planning and post-flood rehabilitation coordination.

In the pre-flood phase, the situation awareness module analyzes incoming data to identify potential risk zones. Based on this information, the optimization module computes the shortest and safest evacuation routes to nearby shelters. The communication module then disseminates alerts through SMS and notification systems, ensuring that residents receive timely warnings and can take appropriate action.

In the post-flood phase, the framework focuses on rehabilitation coordination. An LLM-based module facilitates interaction between affected residents and relevant authorities. The chatbot provides multilingual, context-aware assistance by addressing common queries, such as locating relief supplies, accessing shelters, and applying for aid. This integrated approach enables efficient coordination and supports community recovery efforts.

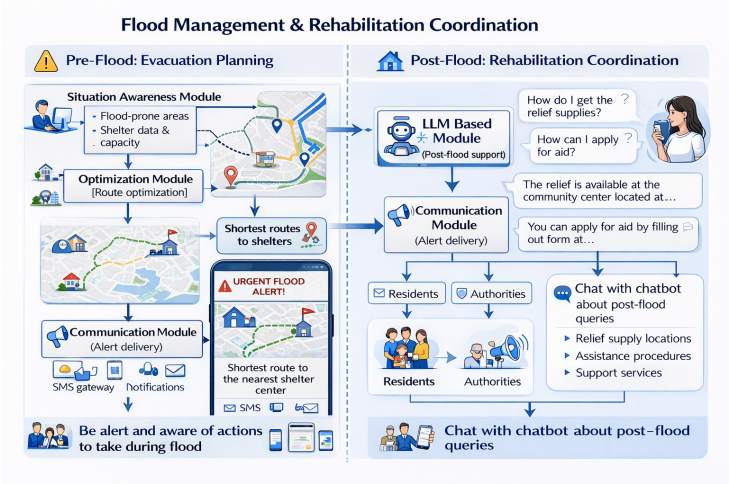


Fig. 1: Flood evacuation planning and rehabilitation

VII. Integrated Approaches: Combining Deep Learning and Agentic AI

Recent surveys highlight the need for integrated systems that leverage the precision of deep learning models and the adaptive intelligence of agentic AI. These hybrid approaches combine data-driven flood prediction (using transformers and GRU) with distributed agentic frameworks responsible for evacuation, resource distribution, and community coordination. The literature suggests that coupling transformer models with GNN-empowered agent systems enables not only accurate predictive analytics but also scalable, decentralized crisis management, making them indispensable for next-generation flood disaster platforms. [12] [7]

The proposed framework [Fig 2] brings together advanced deep learning models and an intelligent agentic AI system to create a unified approach for flood management, evacuation planning, and post-disaster support. It begins by processing rainfall data from Climate Hazards Group InfraRed Precipitation with Stations [CHIRPS] alongside elevation, slope, and land-use information, ensuring that the data is clean, normalized, and ready for analysis. Temporal

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TABLE II: Comparative Performance Analysis: Proposed GRU-Transformer vs. Baseline Models

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Random Baseline	0.5216	0.0961	0.5842	0.1650	0.5501
Logistic Regression	0.8069	0.2799	0.8812	0.4248	0.9215
Naive Bayes	0.7644	0.2356	0.8515	0.3691	0.8505
Support Vector Machine	0.9111	0.4764	1.0000	0.6454	0.9854
K-Nearest Neighbors	0.9639	0.7059	0.9505	0.8101	0.9875
GRU-Transformer (Proposed)	**0.9816**	**0.8305**	**0.9703**	**0.8950**	**0.9974**

rainfall patterns are learned through a GRU network, while a Transformer-based attention mechanism captures important spatial relationships.

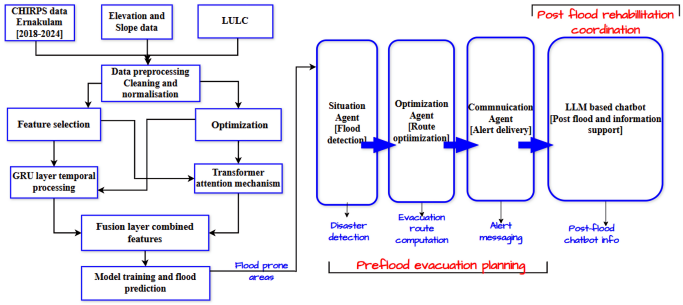


Fig. 2: Proposed Framework

The proposed system utilizes a GRU-Transformer hybrid architecture trained on a multi-modal dataset comprising CHIRPS monthly precipitation (2018–2024), Bhuvan LULC classifications, and high-resolution DEMs for the Ernakulam district. The model processes 12-month temporal sequences using a dual-stream design: a bi-layer GRU (128/64 units) to capture long-term soil saturation trends and a Multi-Head Attention (8 heads) stream to identify localized cloudburst spikes. Optimizing with a 0.0005 learning rate and binary crossentropy loss, the hybrid model achieves a predictive accuracy of 98.16% and a recall of 97.03%. The entire coordination layer is simulated via a LangGraph-based Directed Acyclic Graph (DAG), which orchestrates Situation Awareness, A* Route Optimization, and Communication modules. This modular agentic framework ensures an end-to-end decision-to-alert latency of sub-1.8 seconds, enabling near-instantaneous evacuation planning and autonomous resident notification.

Beyond predictive modeling, the framework implements a dynamic spatial-temporal orchestration layer that bridges the gap between raw hydrological forecasting and actionable disaster response. By ingesting live telemetry from the Open-Meteo API at a granular 1km² resolution, the system continuously updates a district-wide 'Digital Twin' of Ernakulam, merging real-time rainfall accumulation with static geophysical risk factors like topographic slope and soil permeability. The core intelligence lies in its autonomous decision-making backbone, where LangGraph-orchestrated agents execute a constrained A search algorithm* over the OpenStreetMap (OSM) road network to calculate flood-safe evacuation trajectories. This process accounts for dynamic constraints such as historical bridge vulnerabilities and terrain-specific inundation risk, transforming complex model inferences into localized, human-readable evacuation protocols. By integrating RAG-enabled (Retrieval-Augmented Generation) rehabilitation modules, the system ensures that relief coordination is not merely reactive but driven by a continuous loop of high-fidelity environmental feedback and validated disaster management standards

It should be noted that the current evaluation is limited to a regional case study and simulated agent interactions. Further validation under real-world deployment conditions is required to fully assess scalability and robustness.The performance was compared against baseline recurrent models, indicating improved predictive capability. The Table II illustrates the comparative performance analysis of proposed GRU-Transformer vs. baseline models.

VIII. Limitations and Challenges

Despite its promising potential, it has several limitations and challenges. One major limitation is the dependence on the quality and timeliness of input data, including rainfall measurements, geospatial datasets, and infrastructure status [5]. Inaccuracies or delays in these data sources can reduce the precision of flood predictions and evacuation plans. Further, the real-time operational deployment of deep learning models and multi-agent systems demands significant computational resources and efficient data pipelines, which may pose technical and infrastructural constraints, especially in resource-limited settings. The particle swarm optimization (PSO) used for route optimization [21], while powerful, may struggle with scalability in extremely large or complex urban areas featuring dense road networks and variable population dynamics. User compliance and accurate communication during evacuation also present social challenges that AI models alone cannot fully address. Additionally, the integration of multilingual conversational AI for post-disaster rehabilitation requires extensive natural language processing capabilities across dialects and cultural contexts, complicating deployment [9].

IX. Future Scope

The research could explore furthermore by integrating additional data modalities such as satellite imagery, social media feeds, and IoT sensor networks to enhance real-time situational awareness and predictive accuracy. Advances in scalable, decentralized optimization algorithms beyond PSO

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may improve evacuation route computation under increasingly complex conditions. Incorporating behavioral models of evacuee movement and decision-making can refine simulations and improve practical applicability. Enhancing the multilingual abilities of rehabilitation chatbots through continual learning and community-specific customization will further strengthen support services. Finally, piloting the framework in diverse flood-prone regions with feedback loops from emergency managers and affected communities will be essential to validate the system's effectiveness, usability, and equity, ensuring it is responsive to on-the-ground realities.

X. Conclusion

Kerala's unique geographic and climatic conditions make it particularly vulnerable to severe and recurring floods, necessitating innovative, adaptable, and community-centered flood management solutions [1]. The integration of deep learning with agentic AI, especially incorporating multi-agent systems, provides an exciting and highly promising avenue for addressing these challenges. These technologies are well-suited to Kerala's complex hydrological dynamics and diverse population because they allow for real-time, localized flood prediction combined with coordinated evacuation and post-disaster rehabilitation efforts.Using advanced deep learning models such as transformers and GRU networks enables precise and timely forecasting, crucial for issuing early warnings before floods escalate. Agentic AI, empowered by multi-agent systems, facilitates decentralized and dynamic decision-making, making evacuation planning more efficient and equitable. This cooperative model ensures that the unique needs of different communities across Kerala can be met promptly and effectively. In sum, combining deep learning's predictive power with the flexible, intelligent coordination of multi-agent AI systems can revolutionize Kerala's flood disaster management. As demonstrated by ongoing AI-driven disaster initiatives within the state, such integrated approaches not only promise technical accuracy and operational efficiency but also enhance community resilience and inclusivity in managing floods and their aftermath.This research direction is not only timely but critical, offering scalable solutions to save lives, reduce economic losses, and build sustainable flood response frameworks that empower Kerala's people and administration alike.

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